REVISION 2026

Construction Materials Lab

Diploma in Civil Engineering

Code: 2018 | Credits: 1.5 | Semester: 2

Course Outcomes (CO)

- **CO1:** Identify the physical properties of bricks and prepare cement mortar. (Apply)
- **CO2:** Identify soil types and flooring tiles. (Apply)
- **CO3:** Identify special construction materials and determine bulking of sand. (Apply)

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

Module 1: Bricks & Mortar

Expt 1: Dimension & Density of Bricks

Measure dimensions of 10 bricks and find average dimension, weight, and density.

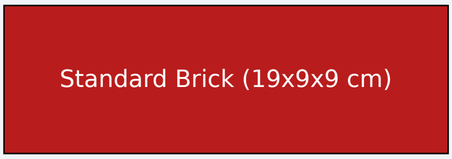
Expt 2: Field Test on Bricks

Perform field tests including dropping, striking, and scratching by nail.

Expt 4: Setting out of Building

Setting out a small building in the field using ropes and pegs.

SVG: Brick Dimension Check



> [illegible] [illegible], [illegible] [illegible] [illegible] [illegible] [illegible]
> [illegible] [illegible] *(Setting out)* [illegible] [illegible].

Module 2: Soil & Flooring

Expt 5: Soil Report

Identify various layers and types of soil in the foundation pit and prepare a report.

Expt 6: Flooring Tiles

Identify different types of flooring tiles such as vitrified, ceramic, glazed, and mosaic tiles.

[illegible].

Module 3: Special Materials & Sand

Expt 8: Bulking of Sand

Determine the increase in volume of sand due to moisture content (Bulking).

[illegible] *(Bulking)* [illegible].

Practice Model Question Paper

Practice Model Paper — not an official REV2026 paper

1. Find the average weight and density of the given 5 bricks.

2. Prepare a cement mortar of 1:4 proportion for the given quantity of sand.

Source Declaration

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